

EXPERIMENT - 11

Name : Sharon Elango

Batch : 3

Roll No.: 62

Inside My-db-app Folder:

#Code for [App.js](#) (Frontend) :

```
import React, { useState, useEffect } from 'react';
import axios from 'axios';

function App() {
  const [users, setUsers] = useState([]);
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');

  // Function to fetch users from your Node.js Backend
  const fetchUsers = async () => {
    try {
      const res = await axios.get('http://localhost:5000/api/users');
      setUsers(res.data);
    } catch (err) {
      console.error("Error fetching data:", err);
    }
  };

  // Run this when the page loads
  useEffect(() => {
    fetchUsers();
  }, []);

  // Function to send new user to your Node.js Backend
  const handleSubmit = async (e) => {
    e.preventDefault();
    try {
      await axios.post('http://localhost:5000/api/users', { name, email
    });
    setName(''); // Clear the input
```

```

    setEmail(''); // Clear the input
    fetchUsers(); // Refresh the list
  } catch (err) {
    alert("Backend not responding. Is your server.js running?");
  }
};

return (
  <div style={{ padding: '40px', fontFamily: 'sans-serif' }}>
    <h1>Experiment 11: MongoDB Integration</h1>

    <form onSubmit={handleSubmit} style={{ marginBottom: '20px' }}>
      <input
        placeholder="Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
        style={{ marginRight: '10px', padding: '8px' }}
      />
      <input
        placeholder="Email"
        value={email}
        onChange={(e) => setEmail(e.target.value)}
        style={{ marginRight: '10px', padding: '8px' }}
      />
      <button type="submit" style={{ padding: '8px 15px', cursor:
'pointer' }}>
        Add to MongoDB
      </button>
    </form>

    <hr />
    <h3>User List from Database:</h3>
    <ul>
      {users.map((user) => (
        <li key={user._id}>
          <strong>{user.name}</strong> - {user.email}
        </li>
      ))}
    </ul>
  </div>

```

```
);  
}  
  
export default App;
```

#Code for [server.js](#) (Backend) :

```
// 1. ADD THESE TWO LINES AT THE VERY TOP TO FIX THE DNS ERROR  
  
const cors = require('cors');  
app.use(cors());  
  
const dns = require('node:dns');  
dns.setServers(['8.8.8.8', '8.8.4.4']);  
  
const express = require('express');  
const mongoose = require('mongoose');  
const cors = require('cors');  
  
const app = express();  
app.use(express.json());  
app.use(cors());  
  
// 2. DATABASE CONNECTION  
// Replace 'SharonPass123' with the NEW simple password you just set in  
Atlas  
const MONGO_URI =  
"mongodb+srv://Sharon:SharonPass123@cluster0.lrgqumk.mongodb.net/Experiment11?retryWrites=true&w=majority";  
mongoose.connect(MONGO_URI)  
  .then(() => console.log("🚀 Connected to MongoDB Atlas Successfully!"))  
  .catch(err => {  
    console.error("❌ Connection error detail:");  
    console.error(err.message);  
  });  
  
// 3. DATA SCHEMA  
const UserSchema = new mongoose.Schema({  
  name: { type: String, required: true },
```

```
    email: { type: String, required: true }
  });

const User = mongoose.model('User', UserSchema);

// 4. REST API ROUTES
app.get('/api/users', async (req, res) => {
  try {
    const users = await User.find();
    res.json(users);
  } catch (err) {
    res.status(500).json({ error: "Failed to fetch users" });
  }
});

app.post('/api/users', async (req, res) => {
  try {
    const newUser = new User(req.body);
    await newUser.save();
    res.status(201).json(newUser);
  } catch (err) {
    res.status(400).json({ error: "Save failed" });
  }
});

const PORT = 5000;
app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});
```

#OUTPUT :

< > 🌐 localhost:3000

Experiment 11: MongoDB Integration

User List from Database:

- **Sharon Elango** - sharonelango10@student.sfit.ac.in
- **Anlea Maria Jose** - anleamjose@student.sfit.ac.in
- **Madison Gonsalves** - madisongonsalves@student.sfit.ac.in
- **Bhumi Koli** - bhumikoli@student.sfit.ac.in
- **Chandrika Jethva** - chandrikajethva@student.sfit.ac.in
- **Valenka Stacey Ravibalan** - valekasravibalan@student.sfit.ac.in